Liver Cirrhosis

الأربعاء Wednesday

يوليه ٢١ July 21

١ صفر ١٤ ابب

- virus A → Acute hepatitis only
- loss of liver func = liver cell failure
- compensated cirrhosis = No failure part is still functioning
- decompensated cirrhosis = failure Jaundice, ascites حمل بقى, Lledema
- virus B more dangerous as it is an oncogenic virus can cause HCC without causing cirrhosis first.
- ∴ Sattt is for life and must be monitored regularly.
- Contacts if have (-) serology must take the vaccine even if they have chronic C

JULY

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Tuesday الثلاثاء

20 July ٣٠ محرم ١٣ أبيب ٢٠ يوليه

- Acute hepatitis B ميدخلش في cirrhosis
لازم يحصل الـ sequence دة
acute → chronic → cirrhosis

- NASH = Metabolic associated steato liver disease = Metabolic associated fatty liver (MASLD) ***
Fatty liver → inflammation → fibrosis → cirrhosis

pt obese, HTN, diabetes, hyperlipidemia having fatty liver
-- /سكر/ ضغط من عندة الدنيا ظبط ولا مخسش
↓
↑↑ liver enzymes why ??
- due to steatohepatitis
Not viral hepatitis (B, C)

- Obese with fatty liver
لازم اطلبلة
- lipid profile
- HbA1c
- thyroid func

يوليه

السبت	الأحد	الاثنين	الثلاثاء	الأربعاء	الخميس	الجمعة
					١	٢
٣	٤	٥	٦	٧	٨	٩
١٠	١١	١٢	١٣	١٤	١٥	١٦
١٧	١٨	١٩	٢٠	٢١	٢٢	٢٣
٢٤/٣١	٢٥	٢٦	٢٧	٢٨	٢٩	٣٠

- any metabolic disturbance ⟶ steatohepatitis

- Steatohepatitis B (reversible) can return to normal liver if we controlled metabolic disturbance present لكن لو دخل في fibrosis خلص كدة

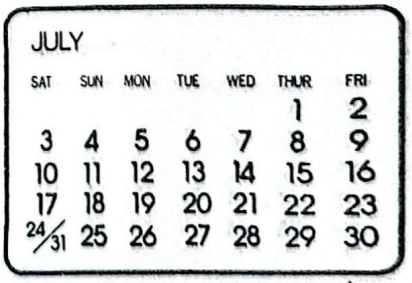
- Fatty liver can end with HCC
- Now causes of cirrhosis
  50% B,C 50% NASH

* Ex:- Causes of cirrhosis:
  - Acute B
  - Acute C
  - Chronic hepatitis ✓

- Primary billiary cirrhosis
- Primary sclerosing cholangitis

Both are cholestatic liver disease cause Jaundice, pruritis, hepatomegaly then cirrhosis why??

= Due to accumulation of bile in biliary channel causing irritation of liver tissue ⟶ fibrosis, cirrhosis



Sunday

18 July

٢٨ محـــرم ١١ أبيـــب

١٧ يوليه

* Cholestatic causes of liver cirrhosis ?

1- Primary billiary "

2- Primary sclerosing "

- Autoimmune hepatitis

Female, hepatomegaly, skin lesion, arthritis

hair loss

بحكون بتكشف مثلاً جلدية / روماتيزم و يطلب

liverenzymes/sonar

↑liver enzymes, hepatomegaly الراقي

Autoimmune liver disease اشك في

Diagnosed by autoimmune marker

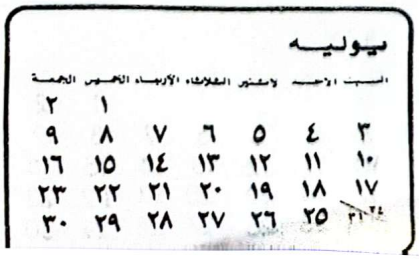
TTT :- steroids, azathioprine

(Immunotherapy)

- Cardiac cirrhosis not common

يعني واحد عنده

Chronic congestive right sided HF



- Hemochromatosis, Wilson, α_1-antitrypsin deff are genetic disorders.

[1] Hemochromatosis (inc intestinal abs of Fe)

الحديد عمال يترسب جوا الliver

(iron overload disease) → oxidative stress → hepatomegaly, joints, bones, cardiac problems, skin pigmentation, Amenorrhea

III- chelation therapy (defuroxamine) and phlebotomy

* Reversible, treatable as long as it didn't develop to cirrhosis

[2] Wilson . Cu ppt in liver, brain

. hepatolenticular degenration

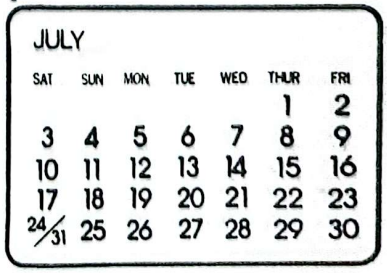
∴ It's a disorder in Cu metabolism

in liver → chronic hepatitis, hepatomegaly

in brain → Extrapyramidal manifestation

. Mostly in female having chronic hepatitis (↑liver enzymes) + gait disturbance as chorea

يتبقى العيلة كلها عندها



TTT :- penicillamine, trientine

3 α_1-anti-trypsin deff is a lung liver disease causing chronic hepatitis, cirhosis

Signs, Symptoms

- اى حد عنده hepatomegaly لازم ادور على السبب

→ Spleenomegaly, Ascites

complication لو حصلوا اعرف انه دخل في

Ex:-

pt with chronic hepatitis and then became cirhotic but pt by general, clinical examination there is no failure (compensated) no complication

TTT :- Supportive therapy as vit E, B12

* Once spleenomegaly, Ascites develop → Complication يبقى حصل

يوليه

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CamScanner

- Ascites may be associated with hydrocele in male (fluid inside scrotum around testicle)
*Any male presenting with this liver cirhosis اسأل في

- Jaundice start in the eye then skin. dark urine ← يعني Jaundice موجود

Acute hepatitis pt first symptom is dark urine as tea indicating non ecteric phase

- Astrexis → Flapping tremors

- Fetor hepaticus (bad breath odour) due to methyl mercaptine impaired metabolism.

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Wednesday الأربعاء

14 July ١٤ يوليه

٢٤ محرم ٧ أبيب

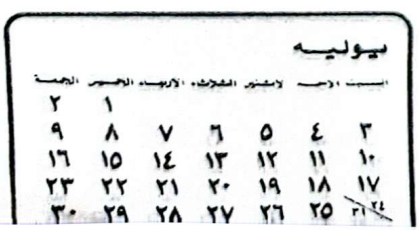
Manifestations of liver cell failure :-
- Jaundice
- Ascites
- Astrexis
- palmar erythema - spider nevi

غالبًا باقي الأعراض كمان

[14, 15]

- Spot diagnosis → Jaundice
- Causes → livercell failure, cholestasis as 2 primary
- Labs needed → serum billirubin

* Spider nevi
- Arteriole surrounded by dilated capillaries due to Nitric oxide (VD) released as result of oxidative stress in liver.
- It's a manifestation of hyperestrogenism that occur in liver cell failure
- Hyperestrogenism → spider nevi
  ↳ female hair distribution



* Spider nevi is normal in female in reproductive period but abnormbal in males, post menopausal female.

spider nevi = spider hemangioma

21 Ascites, Jaundice, feminine hair distribution

22 Localized Ascites

23 Feminine hair distribution, abdominal distention, visible veins

* Normally abdomen flat, concave at sides if there is distention 5F اشك في
- Fat - fluid (ascites)
- fetus - fecal matter

24 Ascites
complication of liver cirrhosis
TT: Diuretics, salt restriction

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Monday الاثنين

12 July ٢٢ محرم ٥ أبيب ١٢ يوليه

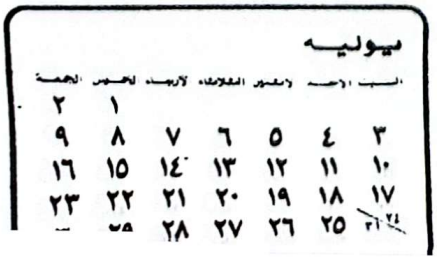
25 LLedema

26 Uedema, cellulitis

28,29 Abdominal distention, feminine hair distribution, abdominal wall vein (Portal HTN), umbulicus abnormal.
Complications of liver cirrhosis

* Spider nevi
feminine hair distribution
Gynecomastia
in Male are due to disturbance of estrogen metabolism in liver cell failure

31 Blackening in Sonar → Fluid



- 2 things in Nails

- Leuconychia → white nails of liver cirrhosis
- Koilonychia → nail spooning in iron deff anemia

Complications

- Spleenomegaly, hyperspleenism → anemia, thrombocytopenia, leucopenia
- Ascites → spontaneous bacterial peritonitis → hepatorenal syndrome (liver, renal failure due to liver not chronic renal disease)
- Hepatic encephalopathy due to hyperammonemia liver can't clear it
- Itching due to hyperbillicubinemia

Hyperbillicubinemia cause both Jaundice and itching

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CamScanner

Saturday السبت

10 July ١٠ يوليه

٢٠ محرم ٣ أبيب

- Liver synthesize vit k dependant factors
In liver cell failure pt may have either thrombocytopenia / hyper coagulable state but most common pt is having bleeding tendancy.

- Portal HTN can result in Ascites (may cause SBP), Bleeding

- Hepato renal more in pts with Ascites having SBP

- Hepato pulmonary, Lung problem due to portal HTN (liver problem)

- Hepatic Osteodystrophy :-
Osteoarthritis, osteoprosis due to liver disease

الجمعة

Diagnosis

Friday

١٩ محرم ٢ ابيب

٩ يوليه

July 9

* Liver enzymes (ALT, AST) are not indicators of liver cell failure but they're indicators of activity (inflammation) لياكلن السبب

* Indicators of liver cell failure :-
  - Serum Albumin not protein (Marker of liver func)
  - Prothrombin

* Finding Cause of Cirhosis :-
  - liver func → albumin → Cirhosis + failure
  - hepatitis marker for B, C if (-) ↓
  - see metabolic profile of pt drugs taken, pt appearance may be hypothyroidism if all normal ↓
  - Autoimmune marker

causes الشي بترتيب اشي

Liver biopsy اخر حاجة تعمل
(Gold standard)

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* Sonar shows cirhosis, ascites, portal HTN, spleenomegaly

* Any pt with cirhosis must have endoscopy as screening to see if he develops portal HTN by presence of varices,
Must be managed at once by ligation (الربط)
to prevent bleeding
∴ So Endoscopy is both diagnostic, therapeutic

* Treated virus C pts must have follow up every 6 Mo by sonar, HCC marker (α-feto protein)

* Also treated virus B pts " " every 6 Mo by renal func, liver enzy, sonar
chronic hepatitis اشوف حو
cirrhosis بس ولد قلب
⊕ HCC marker



* Managment of liver cirhosis :-

Treatment of the cause if treatable / prevent, treat complicaboons